TaxoTox, A Shiny Application for Mixture-Toxicity Assessment of Aquatic Pollutant Assemblages

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Abstract

TaxoTox is an open-source R/Shiny application for rapid assessment of the mixture toxicity of environmental pollutant assemblages. Given a table of measured compound concentrations at one or more water samples, TaxoTox (i) automatically resolves compound names to CAS Registry Numbers (CASRNs) using a curated internal lookup table augmented by optional PubChem REST API queries, (ii) retrieves taxon-specific median acute LC50/EC50 (lethal effect) values from the US EPA ECOTOX Knowledgebase, (iii) computes per-compound Toxic Units (TUs) and summed Pollution Toxicity Indices (PTIs) separately for three functional taxonomic groups; algae, crustaceans, and fish, and (iv) presents results as interactive plots, searchable tables, and a downloadable multi-sheet Excel workbook. TaxoTox makes the TU framework accessible to environmental managers and researchers without requiring programming knowledge and is designed for routine screening of monitoring data at the regional or national scale.

1. Introduction

Environmental monitoring programmes increasingly generate compound-concentration data for tens to hundreds of organic micropollutants (Yusuf et al. 2021) measured across many sampling points in time and space. Most detected concentrations fall well below the acute-effect thresholds for a single compound (Bopp et al. 2019; Yusuf et al. 2021)(Bopp et al. 2019; Yusuf et al. 2021). However, the ecological risk of a complex mixture is not adequately captured by single-compound exceedance calculations; they present cumulative toxic effects . Assessing the toxicity of pollutant mixtures has been a challenge throughout the last century (Käer 1927; Bliss 1939). Several methods are proposed in order to address this challenge.

1.1 Assessing cumulative toxicity using the toxic unit framework

The Toxic Unit (TU) framework (Sprague and Ramsay 1965; Sprague 1970; Könemann 1981) addresses the challenge by referring to the measured concentration of each pollutant in the mix and applying bioassay results (per taxonomic group). Toxic units are calculated by dividing the concentration found in a sample by bioassay results (Equation 1).

$$TU_{i,j} = \frac{C_i}{EC_{i,j}}$$

Equation 1: Calculation of Toxic Units (TU) where i indicates a specific pollutant, j indicates a taxon, and C_i is the concentration of the pollutant that was measured in environmental samples, and EC_j is the effective concentration measured for a specific taxon and pollutant.

The toxic units framework enables the assessment of the combined toxicity of a pollutant mix by summing over the toxicity units of all the compounds in the mix to yield a single dimensionless mixture-toxicity per taxonomic group (Equation 2). The sum of TUs is usually termed the Pollution Toxicity Index (PTI). The convention is that when PTI equals or exceeds one, the mixture may cause acute effects on the receptor taxonomic group; values between 0.1 and 1 are assessed as possessing potential chronic or sub-lethal effects. Usually, PTI is calculated for three functional groups: algae, crustaceans, and fish. The separation between taxonomic groups is important because the rank order of mixture risk frequently differs across taxa: a sample dominated by herbicides may pose negligible risk to fish but acute risk to algae.

$$PTI_j = \sum_{1}^{i} TU_{i,j}$$

Equation 2: summing TU to calculate the potential toxicity index value per taxonomic group.

The Concentration Addition methodology is highly adaptable, and several derivatives of the toxicity unit calculation are used. In some cases, regulatory benchmarks (Backhaus and Faust 2012) replace the denominator in equation 1, hence creating a regulatory compliance metric. In other cases, HC5, the lethal concentration for 5% of species (Belanger et al. 2006) is used as the denominator in equation 1, hence creating a metric that is more adequate for ecosystemic effect and environmental protection. Since most of the laboratory bioassays estimate LC50/EC50, equation 3 is often used to establish the HC5 value, where k is a conversion factor between EC50 and HC5. Usually, k is derived in two ways: when data is rich enough (usually more than 5 bioassays per taxonomic group) k is derived empirically using equation 4, and when fewer than 5 bioassays were conducted for a substance, the k used is the mean of k values for all substances in that taxonomic group. When compounds are data-rich (eg. >5 bioassays per taxonomic group), *k* is derived empirically using equation 4. For data-poor substances, this approach is refined by utilizing an empirically calibrated scaling factor derived from the median HC5 to LC50 ratio of data-rich compounds within each taxonomic group, offering a more scientifically robust estimate of species sensitivity compared to standard fixed Assessment Factors.

$$HC5 = k \times EC_{i,j}$$

Equation 3: estimation of HC5

$$HC5_{data} = \exp(\mu - 1.645 \times \sigma)$$

Equation 4: derivation of the EC50 to HC5 conversion factor for data-rich substances

1.2 Assessing cumulative toxicity using independent action models

A prominent criticism of the Concentration Addition (CA) model is its failure to account for non-additive joint action (Rodea-Palomares et al. 2015; Kar and Leszczynski 2019), as it assumes all mixture components share a similar Mode of Action. This criticism motivated the development of several alternative models. The first being the calculation based on the mode of action (Bliss 1939). This model assumes compounds act independently, each causes an effect according to its own dose-response curve, and the probability of the mixture to cause no effect is the product of the probabilities of each compound to cause no effect. Independent action is considered more appropriate when compounds act through varying mechanisms, which is possibly more appropriate in mixes that are found in the environment (as opposed to single factory effluents). The effect of a single compound (E_{i}) is calculated according to equation 4. To reduce data requirements, equation 4 assumes that the slope of the toxic effect is constant with a value of 1. The mixture effect within the Independent action is the product of all compound toxicities (equation 5, $\mathrm{E}_{\mathrm{mix}}$). Which is bounded between 0 and 1, with increasing toxicity certainty for higher values.

$$E_i = \frac{C_i}{LC50_i + C_i}$$

Equation 4: Calculation of a single compound toxicity within the independent action framework

$$E_{mix} = 1 - \prod_i (1 - E_i)$$

Equation 5: Calculation of the combined toxicity of pollutant mix using the mode of action method.

Concentration addition with mode of action (CAMA) is another form of calculation (Drescher and Boedeker 1995) that also incorporates taxonomic groups. The reasoning behind this method is that, within a group of compounds sharing the same mode of action, CA is the correct model (the joint similarity assumption holds), whereas between pollutant groups that share different modes of action, IA is more appropriate. CAMA exploits this by applying CA within mechanistic groups first, then IA across groups. First, toxicity is calculated per taxonomic group and mode of action as shown in equation 5, resulting in a toxicity units matrix of the number of taxonomic groups times the number of modes of action in the dataset. Then, the toxicity units are grouped by mode of action, and the fractional effect is calculated (equation 7). Finally, the product of modes of action per taxonomic group (E_j) is calculated (equation 8), yielding a number bounded between 0 and 1, where 0 is a non-lethal mixture, and 1 is approaching a certain lethal effect.

$$TU_{MOA(g,j)} = \sum_{i \in g} \frac{C_{ij}}{LC50_i}$$

Equation 6: Calculation of toxicity units in the CAMA model, where g stands for mode of action.

$$E_{MOA(g,j)} = \frac{TU_{MOA(g,j)}}{1 + TU_{MOA(g,j)}}$$

Equation 7: Calculation of the fractional toxic effect per taxonomic group and mode of action.

$$E_{mix,j} = 1 - \prod_{g} (1 - E_{MOA(g,j)})$$

Equation 8: calculation of the combined toxicity of a pollutant mix using the concentration addition with the mode of action model.

1.3 Assessment of toxicity bioassays

A critical bottleneck in performing comprehensive mixture toxicity assessment lies in the lack of acute toxicity endpoints within publicly available databases like the US EPA ECOTOX Knowledgebase (Olker et al. 2022). This data gap, particularly for emerging contaminants (Liang et al. 2025), results in an underestimation of true ecological risk in environmental samples. A quick and possibly dirty solution to this data gap is to integrate predictive toxicity modeling approaches with bioassay results. With the rise of machine learning methods, many models that predict substance toxicity were developed

(Schlender et al. 2023). Methods such as quantitative structure-activity relationships (QSARs) are probably the most frequently employed to generate reliable in silico estimates of missing median effective concentrations (EC50s) and enable routine, large-scale application of mixture assessment (Backhaus and Faust 2012).

Of the different toxicity modelling tools, the USEPA open structure-activity relationship App (OPERA), which is validated to meet OECD regulatory guidelines against 750,000 chemicals on the EPA CompTox Dashboard (Mansouri et al., 2018). OPERA provides an open-source suite of machine-learning QSAR models to complement this data (Williams et al. 2017; Mansouri et al. 2018). OPERA works by taking an input file of QSAR-ready chemical structures, SMILES strings, MOL, or SDF formats, and calculating essential 1D and 2D molecular descriptors. To make predictions, OPERA utilizes a weighted k-nearest neighbor (kNN) algorithm, which estimates the properties of a query chemical based on its similarity to the closest chemicals in the multidimensional descriptor space of the training set (Mansouri et al. 2018).The OPERA output which is easily accessible using the CompTox API (Romano et al. 2022) includes 13 common physicochemical properties and environmental fate endpoints, including LC50 values for specific species of fish and crustaceans.

Despite its conceptual simplicity, routine application of the TU framework is operationally demanding. Multiple chemical naming conventions result in different names for the same pollutant, which requires resolving CAS Registry Numbers, toxicity benchmarks must be retrieved and harmonised across units, and results must be computed, visualised, and exported repeatedly for dozens of sites and sampling dates. This makes the actual assessment of mixture toxicity rare. These challenges also meant that the different mixture toxicity assessment methods were probably never thoroughly compared.

With the growing awareness for pollutants of emerging concern, the need to assess the toxicity of pollutant mixtures grows (Manuguerra et al. 2024; Sudarsan et al. 2024). This research aims to describe TaxoTox (https://yairsuari.shinyapps.io/TaxoTox/), a new publicly available software that, to the best of the authors' knowledge, is the only tool that implements pollutant mixture toxicity assessment methods. In its basic form, it calculates the potential toxicity index using the concentration addition method, but the advanced mode also calculates the potential toxicity index based on HC5 (lethality for 5% of the species), concentration addition potential toxicity index while using national concentration indices as the denominator, toxicity using the mode of action method, and concentration addition with mode of action. In this manuscript, we validate the software's output by reproducing published mixture toxicity values derived from a real-world, reference-grade dataset (Covert et al. 2020). Finally, we compare the toxicity assessment for environmental samples across different calculation methods in order to assess whether different calculation models actually produce substantially different risk assessments.

2. Materials and methods

2.1 TaxoTox Software Architecture and Data Sources

TaxoTox (Figure 1) is implemented as an open-source R Shiny application (Chang et al., 2024) designed for rapid assessment of mixture toxicity. It handles the complete mixture toxicity calculation cycle, which includes gathering toxicity reference data, matching pollutant names to CASRNs, and calculating several mixture toxicity indices.

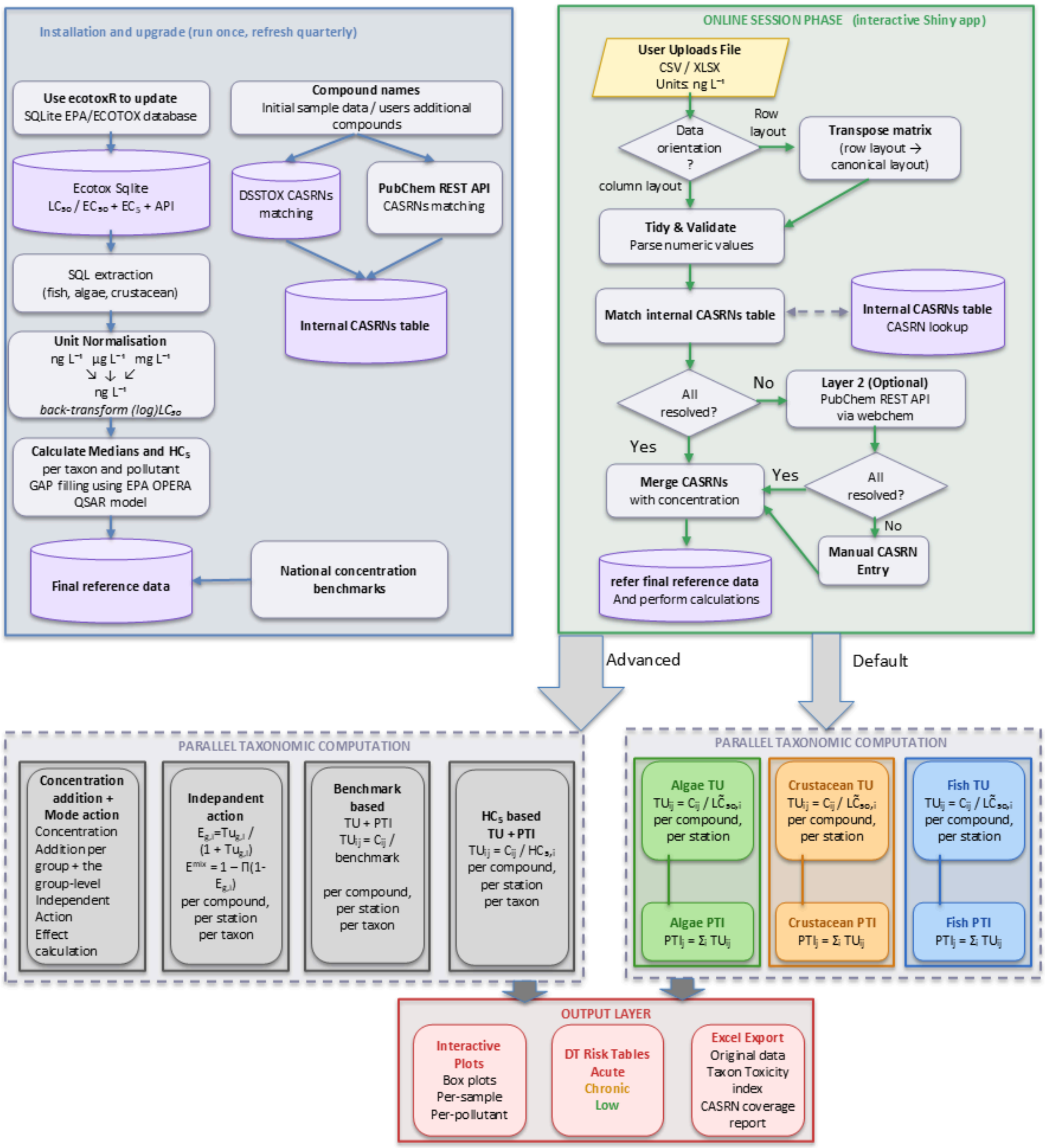

Figure 1: Flow chart describing Mixture Toxicity Assessment of Aquatic Pollutant using TaxoTox.

The TaxoTox implementation consists of two stages: installation/upgrade and toxicity assessment. First, during the initial installation and upgrades, the Compounds table is updated to include compound names (CASRNs) that users have referenced since

TaxoTox was last upgraded. During this stage, the new CASRN entries undergo manual inspection using the PubChem REST API (Kim et al. 2025) to ensure no errors are introduced to the local CAS table. Its worth mentioning that during development, pollutant names from multiple reports of environmental studies and monitoring plans (Abramov 2003; Covert et al. 2020; Topaz et al. 2023, 2024) were associated with CASRNs using EPA's DSSTOX database (Grulke et al. 2019).

Once the CASRNs table was assembled, reference data is merged into TaxoTox tables from the US EPA ecotox knowledge base (Olker et al. 2022) in two stages: 1) all of the LC50 and LD50 tests that were conducted in aquatic conditions for materials (CAS numbers) that were previously used in TaxoTox are queried using the exotox SQL database that is produced by the ecotoxR package (Chang et al. 2024) and than median LC50s and hazardous concentrations (HC5s) are calculated per taxonomic group and CAS number and stored in a TaxoTox tables.

For compounds absent in the ECOTOX database (eg, no toxicity tests were conducted), the toxicity data are complemented by querying the CompTox Chemicals Dashboard CTX REST API (Williams et al. 2017) and extract OPERA-model (Mansouri et al. 2018) predicted LC50 values for fish and crustaceans. Finally, national benchmark data and modes of action are added to the TaxoTox reference table. The final stage of installation is conducting a new deployment of TaxoTox to the R Shiny server, where users can use the app to perform mixture toxicity calculations.

Toxicity assessment is conducted by the user on the TaxoTox Shinyapp website. In short (see the documentation section for additional explanation), the user uploads their raw concentration data as a worksheet, where concentration units must be ng L^{-1}. After the data table is uploaded, TaxoTox attempts to match pollutant names to CASRNs. For compound names that were not identified, the user is suggested a CASRN by using the PubChem API (Kim et al. 2025), and for compounds that were not identified by the API, the user is prompted to enter the CASRN manually. Manual entries are reported to the developers, who merge them with the TaxoTox CASRNs table during upgrades. Once compounds are identified, the user requests to calculate toxicity, using the potential toxicity index as the default. When choosing advanced calculation, the app can calculate

HC5 based PTI calculation, or benchmark hazard index based calculation, or independent action calculation, or concentration addition with independent action calculation.

The output format is composed of plots presenting the most toxic compounds and most toxic samples per taxonomic group, and tables that present TU for each data point. Additionally, the user can download an output Excel workbook that includes the original data, a worksheet for each taxonomic group that displays toxicity units per concentration point and PTI per sample, a report regarding CASRN match, and another report regarding reference data availability for the reported pollutants.

2.1.2 Validation of software calculations

To validate TaxoTox's predicted toxicity index against an independent benchmark, we compared TaxoTox-derived Pesticide Toxicity Index (PTI) scores to the reference PTI values published for the USGS National Water Quality Network (NWQN, Covert et al. 2020). Weekly pesticide concentration data from NAWQA sampling sites (2013–2017) were obtained, with values below the analytical detection limit set to zero rather than treated as detections. These data were processed by TaxoTox to generate predicted PTI scores for the three taxonomic groups. Each TaxoTox output was then merged with the corresponding NWQN reference PTI value, yielding pairs of toxicity scores for the same sample. For each taxonomic group, the top three highest values on each axis were excluded as outliers prior to comparison, and an ordinary least squares regression was fit between the reference PTI and the TaxoTox-predicted PTI, with the resulting slope, intercept, coefficient of determination (R^2), and p-value used to quantify agreement between the two methods. Results were visualized as scatter plots with the fitted regression line and equation annotated, allowing qualitative and quantitative assessment of how closely TaxoTox's predictions track the established NWQN reference values across taxonomic groups.

2.1.3 Software instructions

Datailed

2.1.4 Comparison between mixture toxicity calculation methods

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